HARD · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Problem-Solving and Data Analysis

08

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. 0.99

Choice B is correct. It's given that the density of a certain type of wood is 353 kilograms per cubic meter $\mathrm{kg/m^3}$, and a sample of this type of wood has a mass of $345\ \ \mathrm{kg}$. Let x represent the volume, in m^3, of the sample. It follows that the relationship between the density, mass, and volume of this sample can be written as $\frac{353\ \ \mathrm{kg}}{1\ \ \mathrm{m}^3} = \frac{345\ \ \mathrm{kg}}{x\ \ \mathrm{m}^3}$, or $353 = \frac{345}{x}$. Multiplying both sides of this equation by x yields $353x = 345$. Dividing both sides of this equation by 353 yields $x = \frac{345}{353}$. Therefore, the volume of this sample is $\frac{345}{353}\ \ \mathrm{m}^3$. Since it's given that the sample of this type of wood is a cube, it follows that the length of one edge of this sample can be found using the volume formula for a cube, $V = s^3$, where V represents the volume, in m^3, and s represents the length, in m, of one edge of the cube. Substituting $\frac{345}{353}$for V in this formula yields $\frac{345}{353} = s^3$. Taking the cube root of both sides of this equation yields $\sqrt[3]{\frac{345}{353}} = s$, or $s \approx 0.99$. Therefore, the length of one edge of this sample to the nearest hundredth of a meter is 0.99.

Choices A, C, and D are incorrect and may result from conceptual or calculation errors.

Q2

D

D. 383

Choice D is correct. It's given that a is 230% of b. It follows that $a = \frac{230}{100}b$. It's also given that a is 60% of c. It follows that $a = \frac{60}{100}c$. Since $a = \frac{230}{100}b$ and $a = \frac{60}{100}c$, it follows that $\frac{230}{100}b = \frac{60}{100}c$. Multiplying each side of this equation by $\frac{100}{60}$ yields $\frac{23}{6}b = c$. If c is $p\%$ of b, then $c = \frac{p}{100}b$. It follows that $\frac{23}{6} = \frac{p}{100}$. Multiplying each side of this equation by 100 yields $\frac{2{,}300}{6} = p$. It follows that the value of p is approximately 383.33. Therefore, of the given choices, 383 is closest to the value of p.

Choice A is incorrect. This is closest to the value of p if b is 230% of a, rather than if a is 230% of b, and if b is $p\%$ of c, rather than if c is $p\%$ of b.

Choice B is incorrect. This is closest to the value of p if a is 230% greater than b, rather than 230% of b.

Choice C is incorrect. This is closest to the value of p if c is $p\%$ greater than b, rather than $p\%$ of b.

Q3

D

D. Neither I nor II

Choice D is correct. It's given that data set F consists of 55 integers between 170 and 290 and data set G consists of all the integers in data set F as well as the integer 10. Since the integer 10 is less than all the integers in data set F, the mean of data set G must be less than the mean of data set F. Thus, the mean of data set F isn't less than the mean of data set G. When a data set is in ascending order, the median is between the two middle values when there is an even number of values and the median is the middle value when there is an odd number of values. It follows that the median of data set F is either greater than or equal to the median of data set G. Therefore, the median of data set F isn't less than the median of data set G. Thus, neither the mean nor the median must be less for data set F than for data set G.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4

A

A. 176

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of 0.60 meters. Therefore, the volume of the sample is 0.60^3, or 0.216, cubic meters. It's also given that the sample has a density of 813 kilograms per 1 cubic meter. Therefore, the mass of this sample is $0.216 \text{ cubic meters} \frac{813 \text{ kilograms}}{1 \text{ cubic meter}}$, or 175.608 kilograms. Rounding this mass to the nearest whole number gives 176 kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is 176.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

D

D. 65%

Choice D is correct. Let $n\%$ represent the percent by which the positive quantity x is decreased to result in $0.35x$. The value of n can be found by solving the equation $x - \frac{n}{100}x = 0.35x$. Since x is a common factor of each of the terms on the left-hand side of this equation, the equation can be rewritten as $x1 - \frac{n}{100} = 0.35x$. Dividing each side of this equation by x yields $1 - \frac{n}{100} = 0.35$. Multiplying each side of this equation by 100 yields $100 - n = 35$. Subtracting 100 from each side of this equation yields $-n = -65$. Dividing each side of this equation by -1 yields $n = 65$. Therefore, the expression $0.35x$ represents the result of decreasing the positive quantity x by 65%.

Choice A is incorrect. Decreasing the quantity x by 3.5% yields $x - 0.035x$, or $0.965x$, not $0.35x$.

Choice B is incorrect. Decreasing the quantity x by 35% yields $x - 0.35x$, or $0.65x$, not $0.35x$.

Choice C is incorrect. Decreasing the quantity x by 6.5% yields $x - 0.065x$, or $0.935x$, not $0.35x$.

Q6

D

D. 15.25

Choice D is correct. The mean amount of time that the 20 employees worked for the company is 6.7 years. This means that the total number of years all 20 employees worked for the company is $(6.7)(20) = 134$ years. After the employee left, the mean amount of time that the remaining 19 employees worked for the company is 6.25 years. Therefore, the total number of years all 19 employees worked for the company is $(6.25)(19) = 118.75$ years. It follows that the number of years that the employee who left had worked for the company is $134 - 118.75 = 15.25$ years.

Choice A is incorrect; this is the change in the mean, which isn't the same as the amount of time worked by the employee who left. Choice B is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left and then subtracting the original mean of 6.7 from that result. Choice C is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left.

Q7

30

The correct answer is 30. The situation can be represented by the equation $x(2^4) = 480$, where the 2 represents the fact that the amount of money in the account doubled each year and the 4 represents the fact that there are 4 years between January 1, 2001, and January 1, 2005. Simplifying $x(2^4) = 480$ gives $16x = 480$. Therefore, $x = 30$.

Q8

79

The correct answer is 79. Let x represent the number of zebras in the population in 2014 and let y represent the number of zebras in the population in 2018. It's given that the number of zebras in this population in 2018 was 1.27 times the number of zebras in this population in 2014. It follows that the equation $y = 1.27x$ represents this situation. Dividing both sides of this equation by 1.27 yields $\frac{y}{1.27} = x$, or $\frac{1}{1.27}y = x$. Therefore, the number of zebras in this population in 2014 is $\frac{1}{1.27}$ times the number of zebras in this population in 2018. If the number of zebras in this population in 2014 is $p\%$ of the number of zebras in this population in 2018, then $x = \frac{p}{100}y$. It follows that $\frac{1}{1.27} = \frac{p}{100}$, or $\frac{100}{1.27} = p$, which means p is approximately equal to 78.74. Therefore, the value of p, to the nearest whole number, is 79.

Q9

C

C. II only

Choice C is correct. It's given that the dot plot represents a data set of the number of bursts for 13 eruptions of a steam vent. The median of a data set with an odd number of elements is the middle element when the elements are in numerical order. For 13 elements in numerical order, this is the 7th element. For this data set, the first 4 elements have a value of 1, and the next 5 elements have a value of 2. Thus, the 7th element in the ordered data set is 2 and the median number of bursts for the original data set is 2. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 eruptions, the median of the new data set will be between the 7th and 8th elements in the ordered set, which will also be 2. Therefore, the median number of bursts for the new data set will be the same as the median number of bursts for the original data set. The mean number of bursts for the original data set is found by adding the values of all 13 elements and dividing that sum by the number of elements, 13. Since the data is shown in a dot plot, the sum of the values of the elements can be found by multiplying each element's value by its frequency: $14 + 25 + 32 + 41 + 51$, or 29. Therefore, the mean number of bursts for the original data set is $\frac{29}{13}$. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 bursts, the mean number of bursts for the new data set is $\frac{29+11}{14}$, or $\frac{40}{14}$. Since $\frac{40}{14} > \frac{29}{13}$, the mean number of bursts for the new data set is greater than the mean number of bursts for the original data set. Therefore, of the median number of bursts and the mean number of bursts, only the mean number of bursts is greater for the new data set than for the original data set.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 C

C. $\frac{3x}{17}$

Choice C is correct. It's given that in a scale model, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to 1. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where 1 inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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